INFO4167 – Deep Learning with PyTorch

Food-101 Image Classification Project

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1. Introduction

This project focuses on building a deep learning–based food image classifier using a subset of the Food-101 dataset.

The objective is to classify images into five food categories using a fine-tuned ResNet18 model implemented in PyTorch.

The project includes dataset preparation, augmentation, transfer learning, model training, evaluation, and deployment using a Gradio interface enhanced with Google Gemini Vision.

2. Dataset Overview

The original Food-101 dataset contains 101 food classes with 101,000 images. For this project, a smaller subset was used to enable faster experimentation:

- **5 classes**
- **200 training images per class**
- **60 test images per class**

Selected Classes

- Apple Pie
- Baby Back Ribs
- Baklava
- Beef Carpaccio
- Beef Tartare

Final Dataset Split

- **Training:** 800 images
- **Validation:** 200 images
- **Testing:** 300 images

3. Data Preprocessing & Augmentation

Images were resized to **128×128** and normalized using ImageNet mean and standard deviation.

Training Augmentations

- Random horizontal flip
- Random rotation (15°)
- Color jitter (brightness & contrast)
- Normalization

These augmentations help reduce overfitting and improve generalization.

Validation/Test Transforms

- Resize
- Normalize

4. Model Architecture

A pretrained **ResNet18** model was used with transfer learning.

Modifications

- All layers frozen except the last block of layer4

- Final fully connected layer replaced with:
- Dropout(0.5)
- Linear(512 → 5)

This allows the model to adapt to the five-class classification task while retaining learned ImageNet features.

Total parameters: 11,179,077

5. Training Setup

Hyperparameters

- Optimizer: **AdamW**
- Learning rate: **0.0003**
- Loss function: **CrossEntropyLoss**
- Scheduler: **StepLR(step=5, gamma=0.5)**
- Batch size: **32**
- Max epochs: **15**
- Early stopping patience: **3**

6. Training Process & Early Stopping

Training and validation losses were monitored each epoch. Early stopping was applied to prevent overfitting.

Training Observations

- Training accuracy increased steadily to **96.9%**
- Validation accuracy peaked at **79.5%**
- Early stopping triggered at **epoch 9**

This indicates the model began overfitting after epoch 6–7.

7. Confusion Matrix Analysis

The confusion matrix reveals:

- Apple Pie is sometimes confused with Baklava.

- Baklava is the most frequently misclassified class.
- Beef Carpaccio and Baby Back Ribs show strong separability.

This suggests overlapping visual features between certain dessert-type foods.

8. Sample Predictions

A batch of test images was visualized with predicted vs. true labels.

- Correct predictions shown in **green**
- Incorrect predictions shown in **red**

Most errors occurred between Apple Pie and Baklava, consistent with the confusion matrix.

9. Hyperparameter Summary

- Image Size: 128×128
- Batch Size: 32
- Learning Rate: 0.0001
- Optimizer: AdamW
- LR Scheduler: StepLR (step=5, gamma=0.5)
- Dropout Rate: 0.5
- Epochs Run: 9
- Best Validation Accuracy: 79.5%

10. Deployment with Gradio + Gemini

A user-friendly Gradio interface was built to:

- Upload a food image
- Classify it using the trained PyTorch model
- Generate a description using **Gemini Vision**
- Display confidence scores

Unknown Food Detection

If model confidence < 0.75, the output is labeled **Unknown Food**.

Fallback Descriptions

If Gemini fails to respond, predefined descriptions for each class are used.

This makes the system robust and user-friendly.

11. Conclusion

- This project successfully demonstrates the use of transfer learning with ResNet18 for food image classification.
- Despite using a small subset of the Food-101 dataset, the model achieved 81% test accuracy, showing strong generalization.
- The deployment using Gradio and Gemini Vision enhances usability and provides rich descriptions.